

SOLUTION

The Vacuum Cleaner Problem is solved by allowing the agent to sense the condition of each room and perform appropriate actions until the environment becomes completely clean.

1. Represent the environment as a collection of rooms with either **Clean** or **Dirty** status.
2. Place the vacuum cleaner in an initial room.
3. Check the current room's condition.
4. If the room is dirty, clean it.
5. If the room is already clean, move to the next room.
6. Repeat the process for all rooms in the environment.
7. Continue until every room has been inspected and cleaned.
8. Display the sequence of actions performed and the final state of the environment.